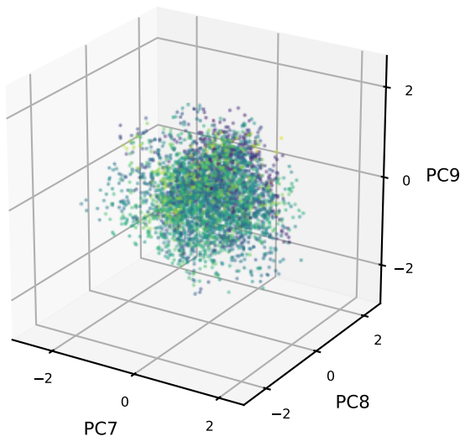
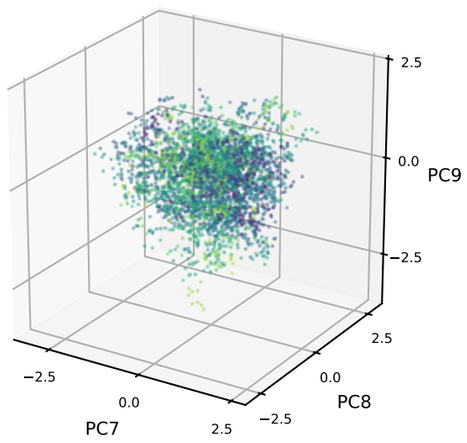
All eight seed pairs · frozen LSTMs at 1B frames · PCs 7–9
3,393 matched states: varying movement, facing, and remembered goals

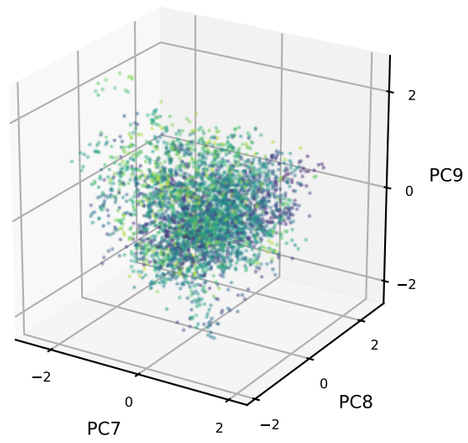
Seed 1 · Grid only
PC7–9: 7.6% of variance



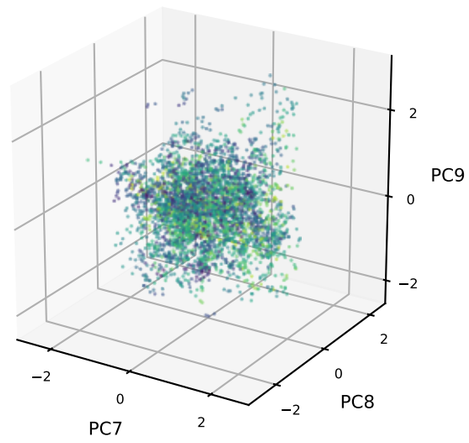
Seed 1 · PGC + grid
PC7–9: 8.0% of variance



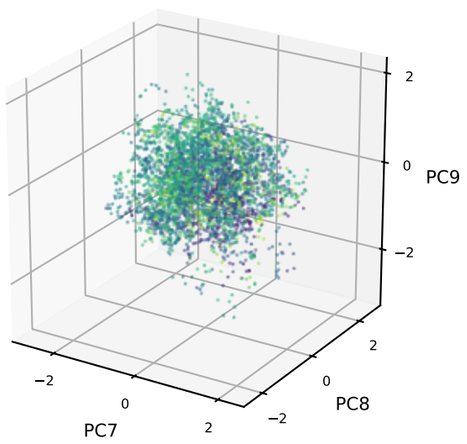
Seed 2 · Grid only
PC7–9: 8.2% of variance



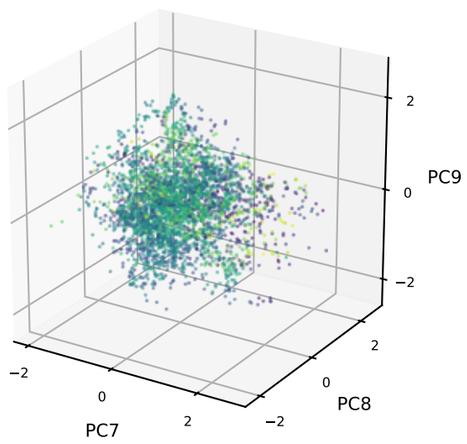
Seed 2 · PGC + grid
PC7–9: 8.6% of variance



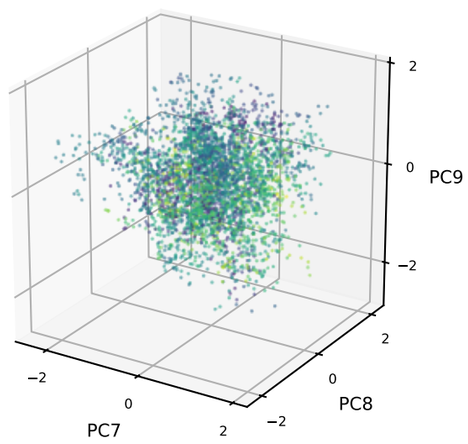
Seed 3 · Grid only
PC7–9: 8.7% of variance



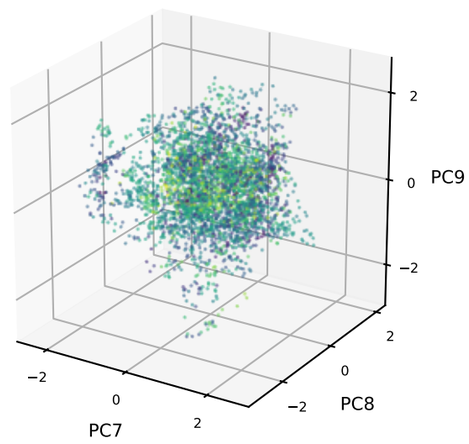
Seed 3 · PGC + grid
PC7–9: 8.0% of variance

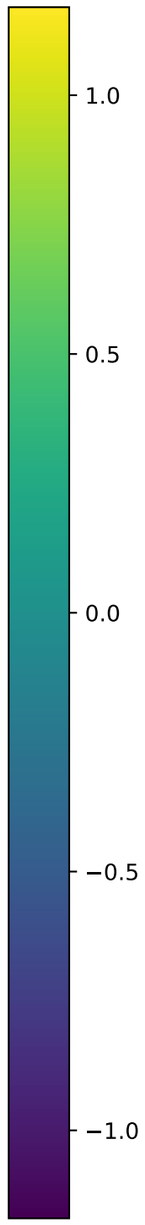


Seed 4 · Grid only
PC7–9: 8.4% of variance

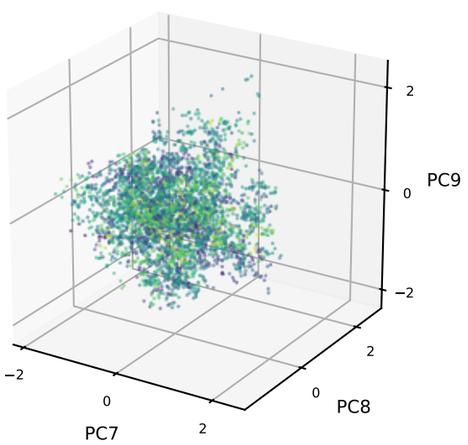


Seed 4 · PGC + grid
PC7–9: 8.0% of variance

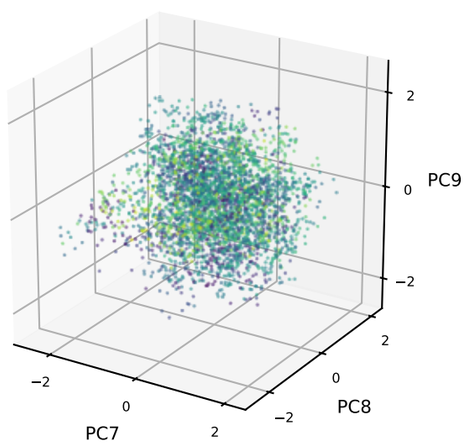




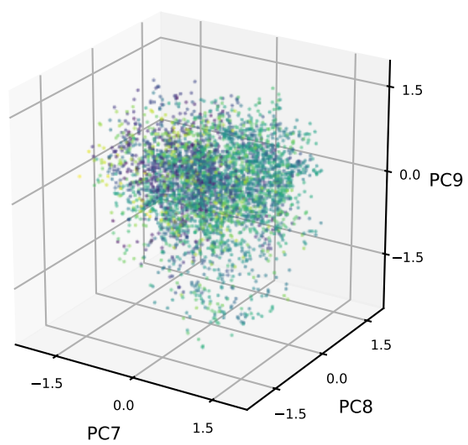
Seed 5 · Grid only
PC7–9: 7.9% of variance



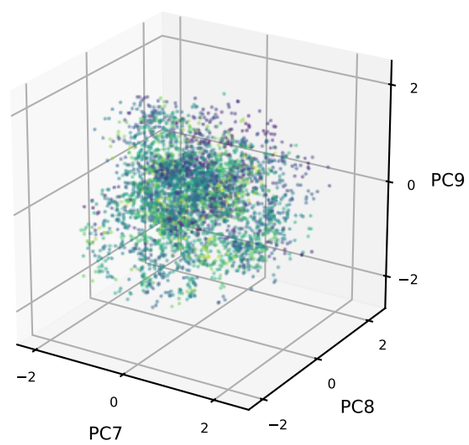
Seed 5 · PGC + grid
PC7–9: 7.9% of variance



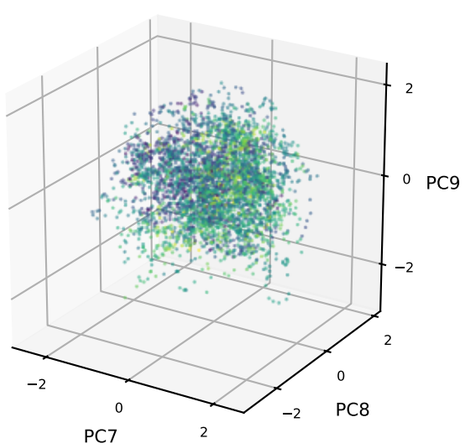
Seed 6 · Grid only
PC7–9: 7.8% of variance



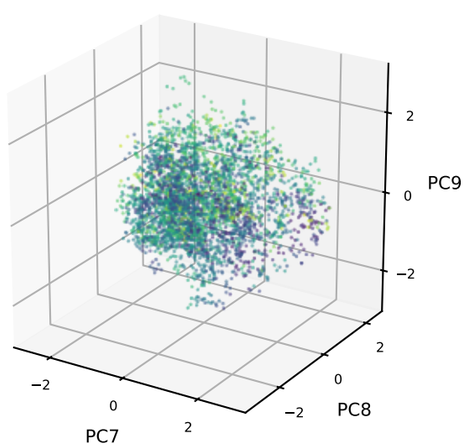
Seed 6 · PGC + grid
PC7–9: 7.8% of variance



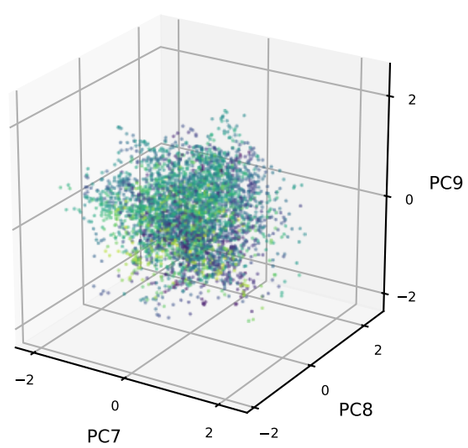
Seed 7 · Grid only
PC7–9: 8.6% of variance



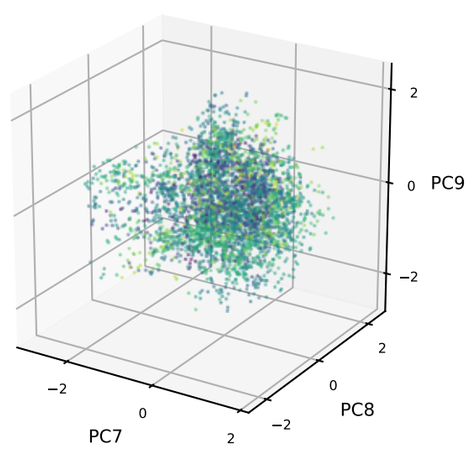
Seed 7 · PGC + grid
PC7–9: 8.5% of variance



Seed 8 · Grid only
PC7–9: 7.8% of variance



Seed 8 · PGC + grid
PC7–9: 8.4% of variance



Independent centered PCA per model; no whitening or UMAP. Equal scale within each 3-D panel; axes are not aligned across models. Shape alone is not a gridness or torus test.